

1. Generalized Coordinate

$$\vec{q} = [q_1, q_2, q_3, q_4, q_5, q_6]^T \in \mathbb{R}^6$$

$$\dot{\vec{q}} \in \mathbb{R}^6, \quad \ddot{\vec{q}} \in \mathbb{R}^6$$

$$\vec{x} = \begin{bmatrix} \vec{q} \\ \dot{\vec{q}} \end{bmatrix} \in \mathbb{R}^{12}$$

* 목표는 trajectory geometry, அதன் $q_{0:N}$ decision variables to kinematic trajectory optimization.

2. Forward Kinematics

$\{B\}$: Base frame

$\{E\}$: End effector frame

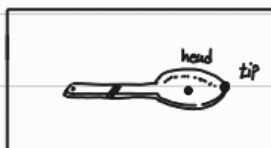
$$T_{BE}(\vec{q}) = \begin{bmatrix} R_{BE}(\vec{q}) & P_E(\vec{q}) \\ 0 & 1 \end{bmatrix} \in SE(3)$$

$$R_{BE}(\vec{q}) \in SO(3), \quad P_E(\vec{q}) \in \mathbb{R}^3$$

• End effectors/ origin + spoon tip/head/ axis + etc.

spoon tip의 local axis $\vec{r}_{tip}^E$

spoon head의 local axis $\vec{r}_{head}^E$



$$\text{World/Ref: } \vec{p}_{tip}(\vec{q}) = P_E(\vec{q}) + R_{BE}(\vec{q}) \vec{r}_{tip}^E$$

$$\vec{p}_{head}(\vec{q}) = P_E(\vec{q}) + R_{BE}(\vec{q}) \vec{r}_{head}^E$$

3. Spoon Orientation model

• local spoon normal $\vec{n}_s^E = [1 \ 0 \ 0]^T$

• local spoon forward $\vec{f}_s^E = [0 \ 1 \ 0]^T$

$$\text{Base frame: } \vec{n}(\vec{q}) = R_{BE}(\vec{q}) \vec{n}_s^E$$

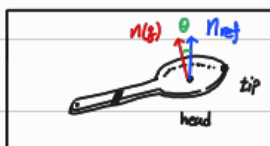
$$\vec{f}(\vec{q}) = R_{BE}(\vec{q}) \vec{f}_s^E$$

Level tilt target: World up vector $\vec{n}_{ref} = [0 \ 0 \ 1]^T$

$$\text{pitch } \theta: \cos \theta = \frac{\vec{n}(\vec{q}) \cdot \vec{n}_{ref}}{\|\vec{n}(\vec{q})\| \|\vec{n}_{ref}\|} = \vec{n}^T(\vec{q}) \vec{n}_{ref}$$

$$= (R_{BE}(\vec{q}) \vec{n}_s^E) \cdot \vec{n}_{ref}$$

→ scalar로 계산

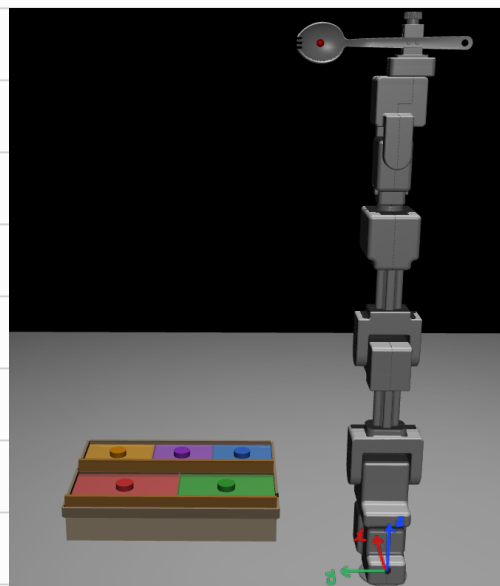


Orientation error: $\vec{e}_n(\vec{q}) = \vec{n}(\vec{q}) \times \vec{n}_{ref}$

$$\begin{aligned} \|\vec{e}_n(\vec{q})\| &= \|\vec{n}(\vec{q})\| \|\vec{n}_{ref}\| \sin \theta \\ &= \sin \theta = \sqrt{1 - (\vec{n}^T(\vec{q}) \vec{n}_{ref})^2} \end{aligned}$$

→ Vector: cross product (pitch의 방향과 크기를 알 수 있음)

$$\Delta \vec{x} = \vec{J}(\vec{q}) \Delta \vec{q}$$



$$e = \begin{bmatrix} P_{ref} - P(q) \\ e_R(q) \end{bmatrix} : \begin{array}{l} \text{Position error} \\ \text{Orientation error} \end{array}$$

$$\rightarrow \Delta q = J^+ e$$

4. head down

$$h(q) = z_E(q) - z_{head}(q) > 0$$

~~Head~~ head down h : $\phi_h(q) = \max(0, h_{min}^{Phase} - h(q))$, $\underline{h_{min}^{Phase}}$: $\nexists$ Phase h head down minimum.

5. Jacobian

$$J(q) = \begin{bmatrix} J_v(q) \\ J_w(q) \end{bmatrix} \in \mathbb{R}^{6 \times 6}$$

spoon tip 的 Jacobian (Position)

$$J_{tip}(q) = \begin{bmatrix} \overline{J_{p,tip}(q)} \\ J_{w,E}(q) \end{bmatrix} \in \mathbb{R}^{6 \times 6}$$

Link 7 的 Jacobian (Angular)

6. Differential kinematics

$$\Delta P_{tip} \approx J_p(q) \Delta q \quad \Delta \theta = J_w(q) \Delta q$$

· DLS (Damped Least Squares) IK (Inverse kinematics)

$$\min_{\Delta q} \|J(q)\Delta q - e(q)\|^2 + \lambda^2 \|\Delta q\|^2$$